Project Title: **Design a module which control SPI flash using I2C(Using SC18IS606 IC)**

Domain: **Electronics**

Project Type: **Schematic/Layout Design**

Project Description:

>To design a module which can control SPI FLASH using I2C interface(Using SC18IS606 IC)

>Perform Pspice simulation for power supply IC's if any(Use TI's IC's only)

>Design a schematic, layout and stack-up using kiCAD EDA tool only. >Use minimum PCB layers and PCB size

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1. Problem Statement and Solution

➢ Problem Statement:

Many microcontrollers have limited internal memory for storing data permanently. In some applications, extra memory is needed to save configuration settings, sensor readings, logs, or other important information.

Although SPI Flash memory provides large storage capacity, some systems mainly use the I²C interface for communication. Connecting SPI memory directly may not always be convenient. Therefore, a simple method is needed to access external Flash memory through an I²C bus.

Need for External Non-Volatile Storage

External non-volatile memory helps increase the storage capacity of a system. Since the data is retained even when power is turned off, it can be used for:

- Storing configuration settings
- Saving sensor data and logs
- Keeping calibration values
- Storing user data and application information

Proposed Solution:

The proposed module provides a bridge between an I²C bus and an SPI Flash memory device. An I²C-to-SPI bridge IC receives commands from the microcontroller through the I²C interface and converts them into SPI commands required by the Flash memory.

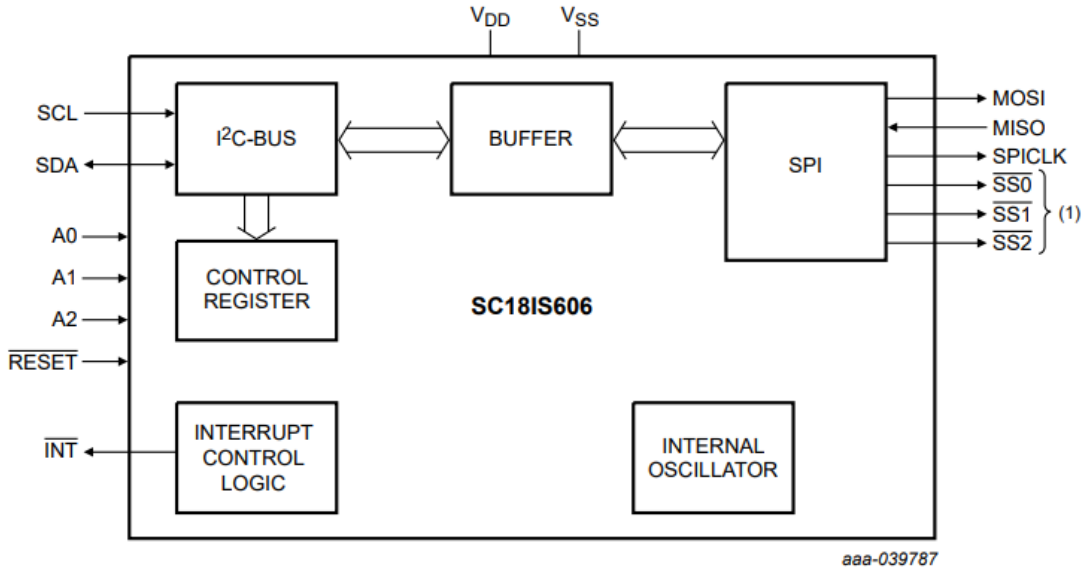
Using this approach, a microcontroller can read and write data to external SPI Flash memory while communicating only through the I²C bus. This makes it possible to add non-volatile storage without requiring a dedicated SPI interface from the microcontroller.

2. SC18IS606 (I2C to SPI bridge IC)

2.1 Introduction:

- SC18IS606 is an I2C to SPI bridge IC developed by NXP semiconductors.
- The IC operates as an I²C slave and SPI master. It means the IC acts as a communication bridge, translating data packets between two different serial communication protocols.
- **I2C Slave Mode**: IC only responds when addressed by the primary host microcontroller (I2C master) using mainly two pins-- **SDA** (Serial Data) and **SCL** (Serial Clock).
- **SPI Master Mode**: Once the IC processes the I²C commands, it switches roles to become the dominant controller on the secondary bus. As the master, it actively generates the **SCLK** (Serial Clock) signal and drives the **CS/SS** (Chip Select) lines. It manages full-duplex data transfer via **MOSI** (Master Out Slave In) and **MISO** (Master In Slave Out) lines, usually running at much higher speeds than the I²C interface.

2.2 Internal Architecture:



1. Unused slave select outputs may be used for GPIO.

Figure 1. Block diagram of SC18IS606

1. I2C-BUS: This block handles all I²C communication. It have two inputs SDA and SCL. It Receives data from microcontroller, detects START and STOP conditions, C\checks device address and reads function ID. Stores received bytes into buffer.

2. BUFFER Block:

- This is temporary memory inside IC.
- It stores data bytes received from the microcontroller through the I²C bus.
- The SC18IS606 contains a 1024-byte internal data buffer.
- After the I2C STOP condition is detected, the buffered data is transferred to the SPI block for SPI communication.
- During SPI transmission, the buffer also stores data received from the SPI slave device through the MISO pin.
- The buffer helps separate I²C communication timing from SPI communication timing, enabling smooth protocol conversion.

3. SPI Block:

- The SPI block operates as an SPI master interface. It converts buffered data into SPI serial communication signals.
- The block generates the **SPI clock** signal (SPICLK) internally.
- **MOSI(Master Out Slave In)** is output line sending data out to peripheral slaves.
- **MISO (Master In Slave Out)** is a input line receiving data back from peripheral slaves.
- The SPI block controls **slave-select** lines SS0, SS1, and SS2 to select SPI slave devices.
- During SPI communication, data is shifted bit-by-bit using internal shift registers and clock generation logic.
- Received SPI data is written back into the internal buffer for later I²C read operations.
- The SPI block supports SPI Modes 0 to 3 and operates up to 1.875 MHz.

4. Control Register: It is a configuration center of the bridge IC. It contains internal registers that store configuration settings sent by the host via I2C. These registers determine parameters like SPI clock speed, SPI clock polarity/phase (SPI modes), and behaviour of slave select lines.

5. Interrupt Control: This controls INT pin. It is used to inform MCU when SPI transfer completed. After SPI transaction finishes, INT becomes LOW, then MCU detects interrupt and reads buffer or clears interrupt.

6. Internal Oscillator: This generates internal timing clock. It provides the core timing clock required for the internal state machines, the Control Register logic, and the generation of the downstream SPICLK frequency without needing an external crystal or resonator.

7. RESET Pin: It resets internal logic, clears states and reloads address pins. When RESET is LOW then buffer is cleared.

2.3 Pin Description:

1. SDA (Serial Data Line): It is the bidirectional data line of the I2C interface.

2. SCL (Serial Clock Line): SCL is the clock input line for I²C communication. It synchronizes data transfer.

3. MOSI (Master Out Slave In): Data stored in the internal buffer is transmitted serially to the SPI slave device through this pin.

4. MISO (Master In Slave Out): MISO is the SPI data input pin. Data returned from the SPI slave device is received through this pin.

5. SPICLK: SPICLK is the SPI clock output generated by bridge IC. It synchronizes SPI data transmission and reception.

6. SS0 / SS1 / SS2: These are SPI slave-select output pins. They are used to select connected SPI slave devices during communication.

7. INT (Interrupt Output): INT is an active-LOW interrupt output pin.

8. RESET: RESET is an active-LOW reset input pin.

9. A0, A1, A2: These pins are used for I2C address selection. These are programmable, different combinations allows multiple IC on the same I2C bus.

10. VDD: VDD is the power supply pin of the IC.

11. VSS: VSS is the ground reference pin.

12. VREFP: It acts as an internal reference connection for proper IC operation.

3. Introduction to Low Dropout Regulator (LDO)

3.1 Introduction:

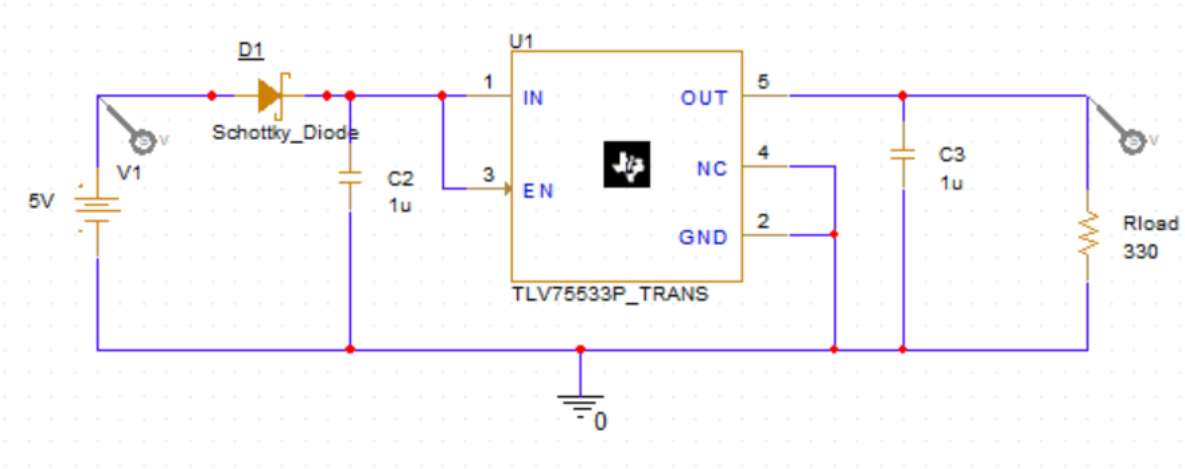
A Low Dropout Regulator (LDO) is a linear voltage regulator designed to provide a stable and regulated output voltage even when the input voltage is only slightly higher than the output voltage.

For SC18IS606 IC (I2C to SPI bridge IC) we need a constant 3.3V operating voltage for reliable communication.

The **TLV75533** low dropout regulator was selected for this project because of the following points :

1. Fixed 3.3V Regulated Output : The TLV75533 provides a fixed and accurate 3.3V output, eliminating the need for external resistor programming ,it simplifies the circuit design.
2. Low Dropout Voltage: In the datasheet of TLV75533 offers a very low dropout voltage of approximately 150mV typical and 212mV maximum for 3.3V output operation at 500mA load current. Conventional LDO like LM1117 provide very high dropout voltage like around 1V.
3. Low Quiescent Current: Quiescent current is the current consumed by IC when it is powered but not performing. The device has a very low quiescent current of approximately 25µA typical. Which helps to improve efficiency of the module we are making.
4. Wide Input Voltage Range: The TLV75533 supports an input voltage range from 1.44V to 5.5V.
5. Small External Component Requirement : The regulator remains stable with only 1µF input and output capacitors.
6. Built-In Protection Features:
   - thermal shutdown,
   - current limiting,
   - undervoltage lockout(Device shutdown itself when voltage drops after safe operational threshold.),
   - foldback current protection(Reduces output current and voltage when short circuit or overload happens).
7. Better Suitability than LM Series Regulators :
   Traditional LM series regulators are older designs with Higher dropout voltage and High power consumption. The TLV75533 is specifically optimized for modern low-power embedded applications operating at 3.3V logic levels.

3.2 Circuit Diagram



The circuit consists of:

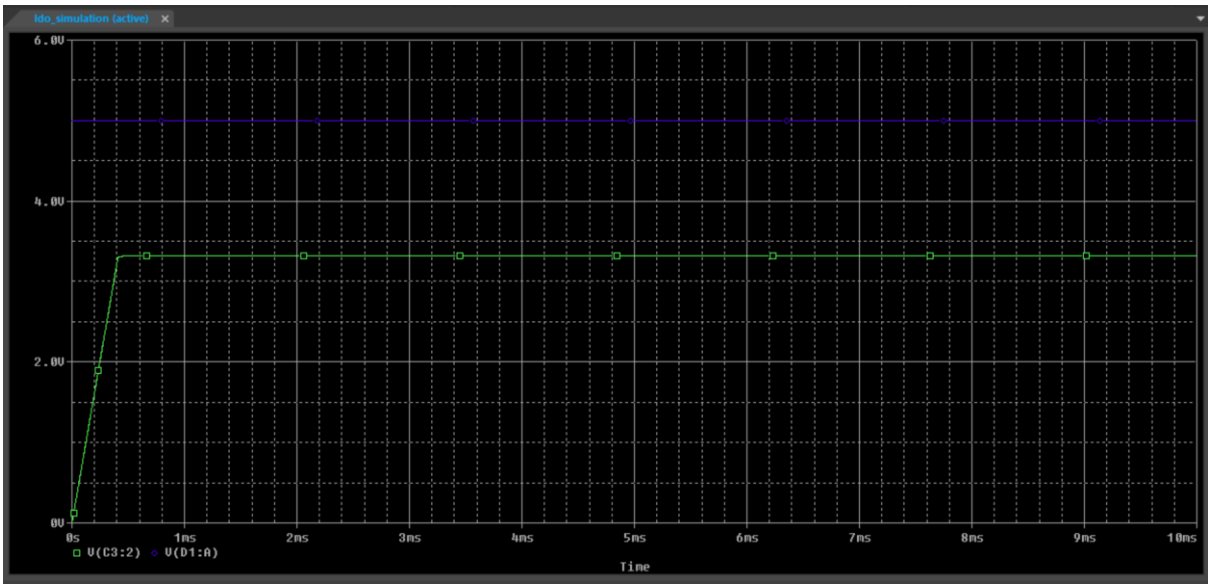
- Input DC supply source
- BAT54 Schottky protection diode
- TLV75533 LDO regulator
- Input and output filter capacitors
- Resistive load for simulation and testing

1. **Input Supply (5V Source)**: 5V dc supply is applied at the input of the circuit . LDO converts 5V to stable 3.3V operational voltage.
2. **Schottky Diode**: It is connected in series with input supply. It is used to give reverse current protection and reverse polarity protection. A Schottky diode is preferred for this module because it have low forward voltage drop, fast switching characteristics, lower power loss compared to ordinary PN junction diodes.
3. **Input Capacitor** : A 1µF capacitor is connected between the input pin and ground according to TLV75533 datasheet.
4. **TLV75533 LDO Regulator** : Main voltage regulator in the circuit which is used to convert higher input voltage to stable 3.3V. It is a 5 pin IC. The EN (enable pin) is connected to input supply. NC(not connected) and GND pins are connected to ground and output in is connected to output capacitor.
5. **Output Capacitor** : A 1µF capacitor is connected at the regulator output. It stabilizes the regulator, reduces output ripple, improves transient response, and prevents oscillations.
6. **Load Resistor**: It represents the practical load connected to the regulator output.

3.3 Transient Analysis of the LDO Circuit

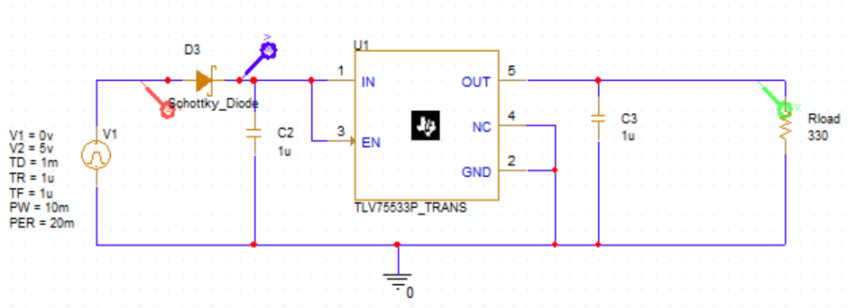
- The startup response of the TLV75533 LDO regulator circuit is observed using time domain transient analysis.
- A 5V input supply was applied to the circuit, and the output voltage variation with time was analyzed.
- The output voltage gradually increased from 0V and settled near the regulated 3.3V output level.
- The regulator achieved stable operation within a short time interval after startup.
- No significant overshoot or oscillations were observed in the output waveform.
- The transient response confirms that the designed power section can provide a reliable and stable 3.3V supply for the I2C-to-SPI bridge IC.

PSPICE Simulation :

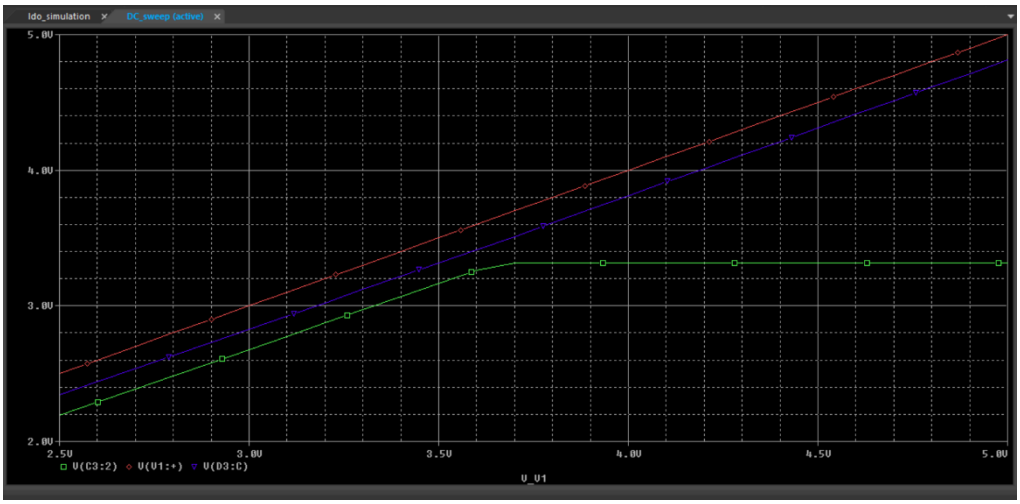


3.4 Line regulation behavior of circuit

Circuit Diagram (with gradually increasing voltage)



DC sweep analysis



- Line regulation analysis is important to ensure if the circuit is providing stable, reliable output voltage despite fluctuations in the input voltage.
- DC sweep analysis is performed to check line regulation.
- Where voltage is varied from 0 to 5V and response of the circuit is observed.
- We can easily calculate **dropout voltage** from this analysis, because we can see above which voltage circuit is giving us stable 3.3V.
- Here I analyzed 3 voltages : Input supply voltage (Red colored, varying from 2.5V to 5V), Voltage after the Schottky diode (actual LDO input), Regulated output voltage of the LDO.
- Initially, when the input voltage is low, the regulator operates in dropout region and the output voltage increases along with the input voltage.
- After the input voltage exceeds the required dropout limit, the regulator enters normal regulation mode and maintains a nearly constant 3.3V output.
- The regulator maintains stable (3.3V) output voltage despite further increase in input voltage, demonstrating good line regulation characteristics.

3.5 Calculations :

1. Dropout Voltage Calculation:

(a) LDO Dropout Voltage:

The dropout voltage of the LDO is calculated using:

$$V_{DO(LDO)} = V_{IN(LDO)} - V_{OUT}$$

From simulation at the point where we start to get stable 3.3V output voltage we get:

Voltage after Schottky diode = 3.5V

Output voltage = 3.3V

Therefore:

$$V_{DO(LDO)} = 3.5V - 3.3V = 0.2V$$

LDO Dropout Voltage is 200mV as expected.

(b)Overall System Dropout Voltage :

Supply voltage = 3.7V

Output voltage = 3.3V

Therefore:

$$V_{DO(system)} = 3.7V - 3.3V = 0.4V$$

So overall system Dropout Voltage is 400mV.

2. Load Current Calculation :

Load resistor used:

$$R_{Load} = 330\Omega$$

Substituting values:

$$I_{Load} = \frac{3.3}{330} = 0.01A$$

3. Power Dissipation in LDO:

In LDO circuit, power is dissipated as heat due to the voltage difference between the input and output.

Power dissipated by the regulator is given by:

$$P_D = (V_{IN} - V_{OUT})I_{OUT}$$

Input voltage after diode = 4.88V

Output voltage = 3.3V

Therefore:

$$P_D = (4.88 - 3.3)(0.01)$$

$$P_D = 0.0158W$$

4. Output Power Calculation:

Output power supplied to the load:

$$P_{OUT} = V_{OUT} I_{OUT}$$

Substituting values:

$$P_{OUT} = 3.3 \times 0.01$$

$$P_{OUT} = 0.033W$$

5. Efficiency :

Efficiency is calculated using:

$$\eta = \frac{P_{OUT}}{P_{IN}} \times 100$$

P_{IN}=4.88×0.01=0.0488W

$$\eta = \frac{0.033}{0.0488} \times 100$$

$$\eta = 67.6\%$$

- **Power dissipation calculation** helps determine how much power is converted into heat inside the LDO regulator. This is important for analyzing thermal performance and ensuring safe operation of the circuit.
- **Efficiency calculation** is used to evaluate how effectively the regulator converts input power into useful output power. It helps compare regulator performance and estimate power loss in the system.
- **Input power (Pin) calculation** helps determine the total power drawn from the supply source. This is useful for estimating overall power consumption of the embedded system.
- **Output power (Pout) calculation** represents the useful power delivered to the load circuit, including the I2C-to-SPI bridge IC and associated components. It helps evaluate whether the regulator is supplying sufficient power for stable operation.

4. W25Q32JV SPI Flash Memory

4.1 Introduction:

- Aim of our module is to access SPI Flash memory using I2C pins of the microcontroller.
- The W25Q32JV SPI Flash memory is used as a non-volatile memory device in the system.
- This IC communicates using SPI protocol. The SC18IS606PW (I2C-to-SPI bridge IC) is used between the microcontroller and flash memory, so that the I2C pins can access the SPI flash memory.

4.2 Specifications of W25Q32JV :

1. Memory Capacity : 32 Mbit (4 MB) Sufficient to store large amounts of configuration data, logs, or firmware images.

2. Operating Voltage : 2.7V to 3.6V Matches the 3.3V rail generated by the TLV75533 regulator, so no level shifting required.

3. The flash memory organises data into pages of 256 bytes for writing and sectors of 4 KB for erasing. This granularity allows the firmware to efficiently manage storage by writing small chunks of data and erasing only the required sector rather than the entire chip.

4. Endurance: 100,000 program/erase cycles Guarantees long-term reliability for repeated data writes during the product's operational life.

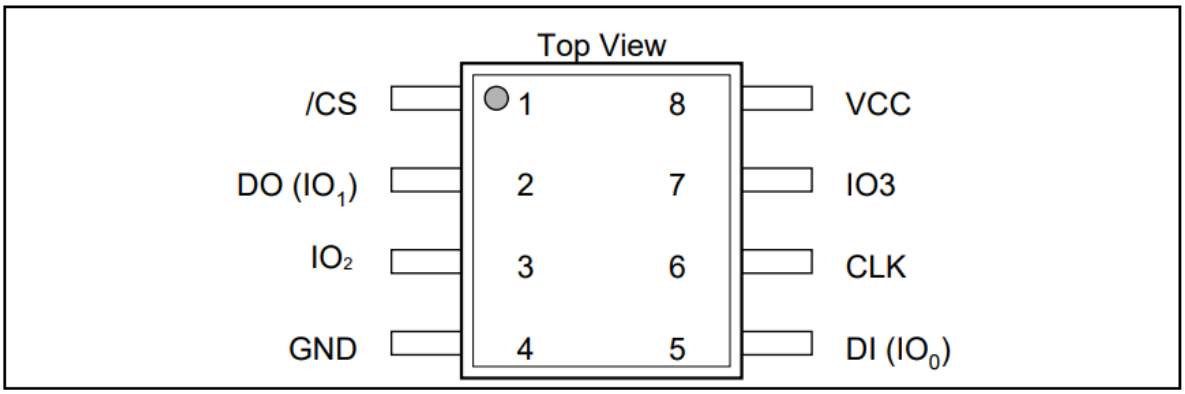
5. Data Retention: the chip's physical structure is robust enough to hold those electrical charges securely for over 20 years.

6. Deep Power-Down (DPD) Mode: This reduces current consumption to under 1 μA during idle periods. Due to this the memory minimizes battery drain when the device is not actively reading or writing data.

7. Operating Temperature: -40°C to +85°C this temperature range ensures working of IC in all conditions.

8. Software and Hardware Write Protection: Protects stored data from accidental corruption, which can be controlled through the SC18IS606PW bridge by the MCU.

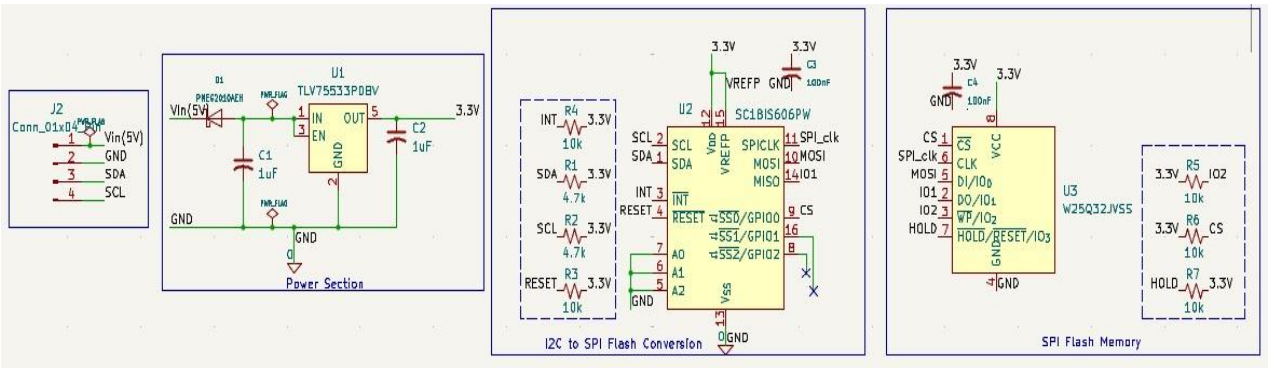
4.3 PIN Configuration:



1. **CS(Chip Select):** Enables the flash memory device during SPI communication.
2. **DO (IO1):** Sends data from flash memory to the master device (MISO line).
3. **IO2:** Used as Write Protect (WP) pin in standard SPI mode.
4. **GND:** Provides ground reference for the IC operation.
5. **DI (IO0):** Receives data from the master device (MOSI line).
6. **CLK:** Synchronizes SPI data transfer using clock pulses.
7. **IO3:** Used as HOLD/RESET pin in standard SPI mode.
8. **VCC:** Supplies operating power to the flash memory IC.

5. Schematic Diagram Overview

5.1 Schematic diagram of Module:



1. Power Section: This section converts 5V input to a stable 3.3V supply voltage for main IC.

#Components used:

a) TLV75533DBV is mainly used to convert 5V to 3.3V.

b) PMEG2010AEH Schottky diode: It used at the input for reverse polarity protection. The diode is rated for 20V and 1A, which is sufficient for low-power embedded applications.

c) Input and Output capacitors: The input capacitor filters supply noise and ripple from the external source. The output capacitor stabilizes the 3.3V output of the regulator.

2. I2C to SPI Flash Controller Section:

This section converts I2C commands from the host controller into SPI communication for accessing the SPI flash memory.

#Components used:

a) **SC18IS606PW (Main Bridge IC):**

It allows a microcontroller with only an I2C interface to communicate with SPI devices.

b) **SDA and SCL Pull-up Resistors (R1 and R2 - 4.7kΩ):**

I2C communication uses **open-drain** outputs. therefore pull-up resistors are mandatory.

R1 and R2 pull the SDA and SCL lines to 3.3V to establish valid **HIGH logic levels**.

c) **RESET Pin Pull-Up Resistor:**

Reset pin is an active-LOW open-drain pin. A **10k Ω resistor** is applied to maintain a stable logic-high state during normal operation.

d) INT Pin Pull-Up Resistor:

The INT pin is an active-LOW open-drain interrupt output. The **10kΩ** pull-up to 3.3V is required to define a valid logic HIGH when interrupt is not active.

e) Address Pin Connections (A0, A1, A2):

The address selection pins are connected to GND. This sets the default I2C slave address of the SC18IS606.

f) Decoupling Capacitor (C3 -100nF):

A 100nF capacitor is connected between VDD and GND near the IC power pins. It suppresses high-frequency switching noise and improves power stability during digital switching operations.

g) SPI Signal Connections:

SPICLK generates the SPI clock signal.

MOSI transfers data from the bridge IC to the flash memory.

MISO receives data from the flash memory.

CS is used to enable or disable the SPI flash device during communication.

3. SPI Flash Memory Section:

This section contains the W25Q32JVSS SPI Flash memory IC used for non-volatile data storage.

#Components used:

a) W25Q32JVSS SPI Flash Memory:

It is a 32M-bit serial NOR Flash memory device from Winbond. It is optimised for systems with limited space, pins, and power.

b) SPI Communication Pins:

CLK receives the serial clock signal generated by the SC18IS606.

DI (MOSI) receives command and data inputs from the bridge IC.

DO (MISO) sends stored data back to the bridge IC.

CS (Chip Select) enables the flash device during SPI communication.

Communication occurs only when the CS pin is driven LOW.

c) Decoupling Capacitor (C4-100nF):

A 100nF capacitor is connected between VCC and GND close to the IC. It reduces

high-frequency supply noise and stabilizes the power rail during memory operations.

d) CS Pull-up Resistor (R5 - 10kΩ):

The CS pin is connected to the slave select output of the SC18IS606.

According to the datasheet, the bridge IC automatically pulls the CS line LOW during SPI data transmission and releases it HIGH after communication is completed.

The external **10kΩ pull-up resistor** ensures that the flash memory remains deselected during power-up and idle conditions.

e) WP Pull-up Resistor (R6 - 10kΩ):

The WP (Write Protect) pin is pulled HIGH to disable hardware write protection. This allows normal write and erase operations of the flash memory.

f) HOLD Pull-up Resistor (R7 -10kΩ):

The HOLD pin must remain HIGH during normal SPI operation. So it is pulled HIGH to disable the communication hold feature. This prevents accidental suspension of SPI communication due to noise or floating inputs.

- The schematic includes all the required components for proper operation of the SC18IS606 bridge IC and W25Q32 flash memory. Voltage regulation, pull-up resistors, and decoupling capacitors have been used to ensure stable power supply and reliable communication. The design follows the recommendations given in the component datasheets and provides a strong foundation for the complete module.

6. PCB Design and Layout

6.1 Footprint Selection:

According to ease of soldering and compactness I selected appropriate footprints. Footprints of ICs (SC18IS606PW, W25Q32JVSS, TLV75533 LDO) were selected according to specifications in their datasheets.

Component	Footprint Used	Reason for Selection
TLV75533 LDO	SOT-23-5	Compact regulator package
SC18IS606PW	TSSOP-16 (0.65 mm pitch)	Matches datasheet recommendation and provides sufficient routing space.
W25Q32JVSS Flash Memory	SOIC-8	Easier soldering and debugging and also matches datasheet recommendations.
Resistors and Capacitors	0603 SMD	Good balance between compact size and manual solderability
Input Connector	1×4 Pin Header (2.54 mm)	Simple interfacing with external microcontroller or breadboard
Schottky Diode	SOD-123F	Compact high-current Schottky diode footprint

Symbol : Footprint Assignments

1	C1	1uF	Capacitor_SMD:C_0603_1608Metric
2	C2	1uF	Capacitor_SMD:C_0603_1608Metric
3	C3	100nF	Capacitor_SMD:C_0603_1608Metric
4	C4	100nF	Capacitor_SMD:C_0603_1608Metric
5	D1	PMEG2010AEH	Diode_SMD:D_SOD-123F
6	J2	Conn_01x04_Pin	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_
7	R1	4.7k	Resistor_SMD:R_0603_1608Metric
8	R2	4.7k	Resistor_SMD:R_0603_1608Metric
9	R3	10k	Resistor_SMD:R_0603_1608Metric
10	R4	10k	Resistor_SMD:R_0603_1608Metric
11	R5	10k	Resistor_SMD:R_0603_1608Metric
12	R6	10k	Resistor_SMD:R_0603_1608Metric
13	R7	10k	Resistor_SMD:R_0603_1608Metric
14	U1	TLV75533PDBV	Package_TO_SOT_SMD:SOT-23-5
15	U2	SC18IS606PW	Package_SO:TSSOP-16_4.4x5mm_P0.65mm
16	U3	W25Q32JVSS	Package_SO:SOIC-8_5.3x5.3mm_P1.27mm

6.2 **Component Placement:**

Component placement was performed by considering signal flow, routing simplicity, and power integrity.

- The SC18IS606 I2C-to-SPI bridge IC was placed near the centre of the PCB since it is the main controller.
- The SPI flash memory was positioned close to the SC18IS606 to minimize SPI trace length.
- The LDO regulator was placed near the power input connector to ensure stable voltage conversion before distribution.
- Pull-up resistors for I2C lines were placed close to the controller pins.
- Decoupling capacitors were placed immediately near power pins of ICs

6.3 Track Width Selection:

Wider traces were used for power lines to reduce voltage drop and improve current handling. Track width of 0.5mm is used for power tracks.

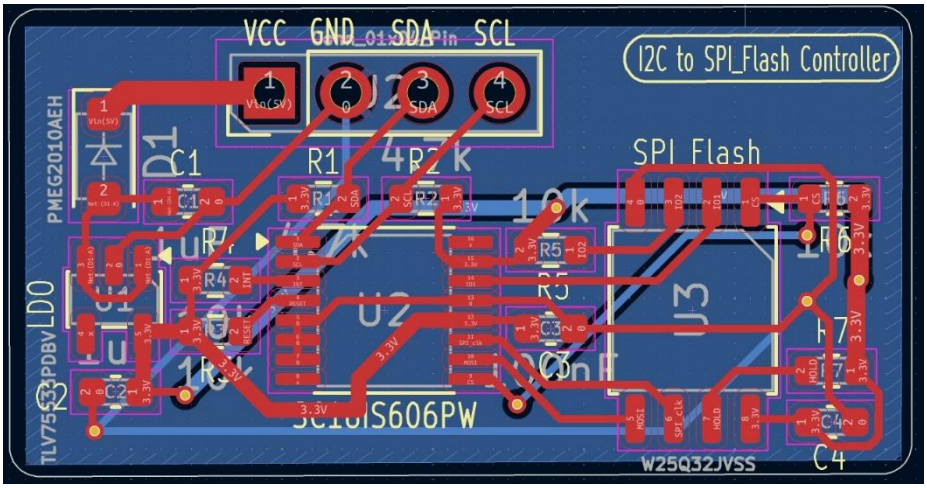
Track width of 0.25mm is given to SPI and I2C tracks for better signal integrity.

6.4 Ground Routing:

A ground copper pour was used across the PCB to provide a common low-impedance return path.

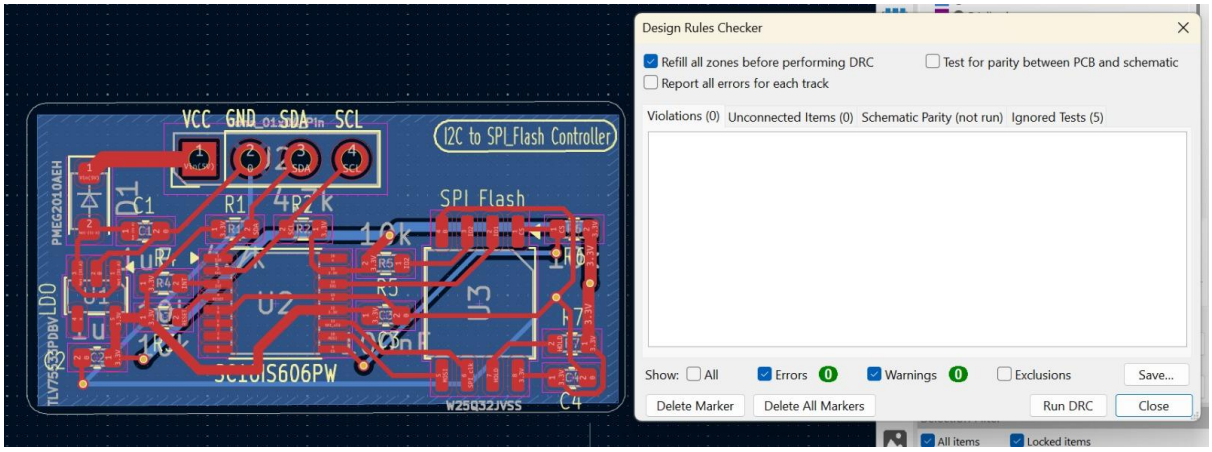
Bottom-layer copper fill was connected to GND net.

Multiple ground vias were added near major components.

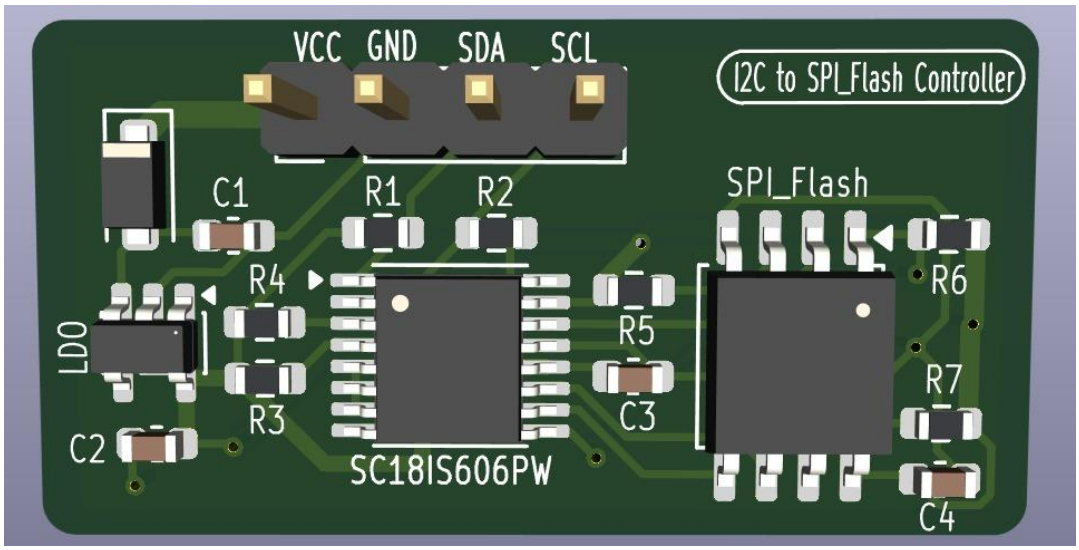


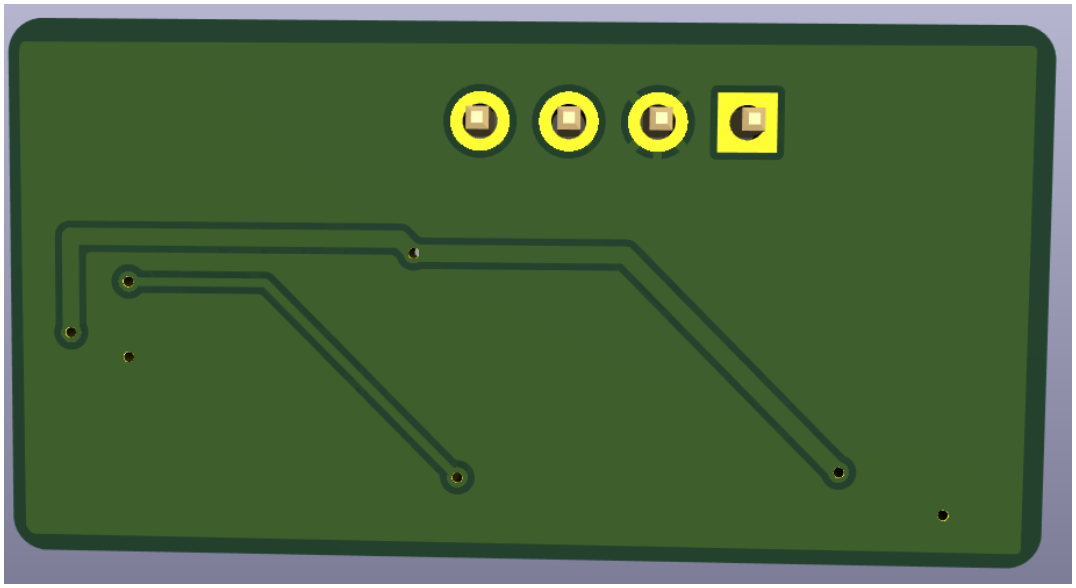
PCB Routing

6.5 Design and Rules Check:



6.6 Final 3D View of PCB:





Working of the Complete Module

- The designed module acts as an interface between an I²C master device (microcontroller) and an SPI Flash memory. The module uses the SC18IS606 I²C-to-SPI bridge to translate I²C commands into SPI transactions, allowing an I²C-only system to access SPI Flash memory.
- **Data Flow:**
  MCU -> I²C Bus -> SC18IS606 -> SPI Bus -> W25Q32JV Flash Memory
- **Step 1: MCU Generates an I²C Transaction:**
  When the microcontroller wants to store data, it places the following information on the I²C bus:
  - SC18IS606 slave address
  - SPI command to be executed
  - Flash memory command
  - Memory address
  - User data

These bytes are transmitted through the SDA line while the SCL line provides clock synchronization. The SC18IS606 receives all transmitted bytes and stores them in its internal buffer.

- **Step 2: SC18IS606 Interprets the I²C Data:**
  The SC18IS606 continuously monitors the I²C bus.
  After receiving a valid I²C packet addressed to it:
  - It decodes the received bytes.
  - Determines SPI slave select output. SS0 is selected according to module.
  - Loads the received bytes into its SPI transmit buffer.
  - Prepares the SPI engine for transmission

  At this point, the information still exists only inside the SC18IS606.
- **Step 3: Conversion from I²C to SPI:**
  The SC18IS606 automatically generates SPI signals:
  - CS → Selects Flash Memory
  - SCLK → SPI Clock
  - MOSI → Data from bridge to flash
  - MISO → Data from flash to bridge

The bridge pulls CS low and begins clocking data onto the MOSI line.

- **Step 4: SPI Command Generation:**

  The SC18IS606 asserts the chip-select (CS) line and generates SPI clock pulses.

  The SPI Flash receives:

  - Write Enable (06h) command
  - Page Program (02h) command
  - Target memory address
  - Data bytes to be stored

- **Step 5: Flash Programming:**

  The W25Q32JV stores the received data into its non-volatile memory array.

  During programming, the flash memory internally performs the write cycle and updates the selected memory page.

- **Step 4: Completion**

  After programming is complete, the flash clears its BUSY status bit and becomes ready for the next operation.

Reading Stored Data Back:

- **Step 1: Read Request:**

  The microcontroller sends a read request to the SC18IS606 via I²C.

- **Step 2: SPI Read Transaction:**

  The SC18IS606 generates an SPI transaction and sends:

  - Read Data command (03h)
  - Target memory address

  to the W25Q32JV Flash memory.

- **Step 3: Data Retrieval:**

  The flash memory outputs the stored data through its DO (MISO) pin.

- **Step 4: Data Transfer to MCU:**

  The SC18IS606 receives the SPI data and places it in its internal receive buffer.

  The microcontroller then reads this data through the I²C interface.

  Thus, data flows in the reverse direction:

  Flash Memory -> SPI -> SC18IS606 -> I2C -> MCU

Erase Operation:

Erase operations are used to clear memory locations before new data is programmed.

- **Step 1: Erase Command Request:**

  The SC18IS606 sends**:**

  - Write Enable (06h)
  - Sector Erase (20h) or Block Erase commands to the flash memory.

- **Step 2: Internal Erase Process:**

  The W25Q32JV internally removes stored charge from the selected memory cells, resetting them to the erased state (logic '1').

- **Step 3: Status Monitoring:**

  The BUSY bit in the Status Register remains set while data erasing is in progress. The microcontroller can periodically read the status register through the SC18IS606 to determine when the operation has completed**.**

Conclusion and Learning Outcomes

The objective of this project was to design and develop an I²C to SPI Flash Memory Interface Module using the SC18IS606 I²C-to-SPI bridge and the W25Q32JV SPI Flash memory. The designed module enables a microcontroller with only an I²C interface to communicate with and utilize SPI-based non-volatile memory.

This project demonstrates the practical implementation of protocol conversion techniques and highlights the integration of multiple communication standards within a single hardware system.

Learning Outcomes

Through this project, the following concepts and skills were learned:

- Understanding of I²C and SPI communication protocols and their differences.
- Study of protocol conversion using the SC18IS606 I²C-to-SPI bridge.
- Understanding the operation of SPI Flash memories and non-volatile data storage.
- Selection of electronic components based on datasheet specifications.
- Design and verification of electronic schematics.
- PCB design using KiCad, including footprint selection, component placement, routing, and grounding techniques.
- Use of pull-up resistors, decoupling capacitors, and voltage regulation circuits in practical hardware design.
- Interpretation and application of manufacturer datasheets during circuit development.

Reference Datasheets:

1. NXP Semiconductors, *SC18IS606 I²C-Bus to SPI Bridge Datasheet*.

2. Winbond Electronics, *W25Q32JV 32-Mbit Serial Flash Memory Datasheet*.

3. Texas Instruments, *TLV755P / TLV75533 Low-Dropout Voltage Regulator Datasheet*.